CV

Basics

Name Luke Lowery

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Website https://lukelowry.github.io

Summary Currently employed at Texas A&M University. EE PhD Student of Power Systems

Work

2024 - Present

Research Assistant

Texas A&M University

- Develops Phasor & EMT models to accelerate the integration of renewables in modern transmission systems.
- Performs novel research on the impact of Geomagnetic Disturbances on the nations electric grid.
- Applies modern mathematics to solve open problems in energy and power research.

2023.05 - 2023.08

High Voltage Intern

SpaceX

- Switchgear & transformer installation, design, and coordination for mission-critical loads (5kV & 35kV)
- Responsible for differential and over-current protection on critical feeder lines to production and launch-pad.
- Designed & implemented 500 kW feeder and building distribution to a new production building in 4 weeks.
- Co-designed the microgrid controller functionality with Tesla Engineers. Functionality allowed our Mega-Packs (smart battery storage) to create seamless transitions from the Utility to isolated island operation.

2022.08 - 2023.01

Power Systems Cooperative Education

SpaceX

- Coordinated with other SpaceX production and launch sites to create a standardized SCADA system. Performed transient analysis of synchronization anomalies to discover coordination errors.
- Introduced solar panel expansion to decrease diesel use, turning Starbase into a 50% green energy site.
- Designed a site-wide SCADA system to monitor and automate the site microgrid using Ignition as the front end.

2022.05 - 2022.08

SCADA Intern

SpaceX

- Automated various power distribution tasks such as load-shedding, interactive fuel-consumption projector, and predictive site-load calculator that is still utilized during launch operations.
- Deployed numerous PLCs, control panels, and sensors to increase grid visibility and control using Allen Bradley, Siemens, and SEL hardware. Communication protocols used: MQTT, DNP, Modbus, Profinet, GOOSE.

2021.05 - 2021.08

Control Systems Intern II

ICS Texas

- Designed and implemented control systems for numerous applications including natural gas sites, boiler systems, and production facilities. Primarily completed with Allen Bradley PLC hardware.

2020.05 - 2020.08

Control Systems Intern I

ICS Texas

- Mastered AutoCAD for the design of control panels that were manufactured in-house. Developed an automated AutoCAD program in LISP to synthesize to accelerate manufacturing.

Education

2023 - 2027

College Station, TX

PhD

Texas A&M University

Electrical Engineering

2019 - 2023

College Station, TX

Bachelor of Science

Texas A&M University

Electrical Engineering

Leadership

2025.05 - Present

Senator

Graduate and Professional Student Government

Student Senator for the ECEN Graduate Department

- Draft legislation that upholds university values and improves the Aggie experience.
- Vote and represent the Department of Electrical Engineering on University policies and procedures.

Publications

2026

System Frequency Nadir and Trajectory Prediction With Discontinuity Constraints in Governor Dynamics

Jongoh Baek, *Luke Lowery*, and Adam B. Birchfield

IEEE Open Access Journal of Power and Energy, 13, pp. 274–285

DOI: 10.1109/OAJPE.2026.3675094

2026

Spectral Graph Filters for Large Power Systems

Luke Lowery and Adam B. Birchfield

2026 IEEE Texas Power and Energy Conference (TPEC), pp. 1–6

DOI: 10.1109/TPEC67884.2026.11513110

2026

Using Spectral Graph Wavelets to Analyze Large Power System Oscillation Modes

Luke Lowery, Jongoh Baek, and Adam Birchfield

59th Hawaii International Conference on System Sciences (HICSS)

2025

EMT Simulation with Spectral Graph Wavelets

Luke Lowery and Adam B Birchfield

2025 IEEE Texas Power and Energy Conference (TPEC), pp. 1–6

2025

Voltage Stability in Plausible Non-Uniform Geomagnetic Disturbances

Luke Lowery and Adam B. Birchfield

IEEE Transactions on Power Systems, pp. 1-12

DOI: 10.1109/TPWRS.2025.3588826

2024

Voltage Stability Interface Limits in Non-Uniform Geomagnetic Disturbances

Luke Lowery and Adam B Birchfield

2024 56th North American Power Symposium (NAPS), pp. 1–6

Presentations

2024.03.01
College Station, Texas

Graduate Speaker
ECEN Townhall

2024.03.01
College Station, Texas

Graduate Speaker
ECEN Fellowship Banquet

2024.03.01
College Station, Texas

Speaker
Modern Methods in Electromagnetic Transient Simulations

2024.03.01
College Station, Texas

Speaker
Voltage Stability in Non-Uniform Geomagnetic Disturbances

Awards

2025.12.01

Best Paper
59th Hawaii International Conference on System Science
Best Paper in Electric Energy Systems Track

2025.11.01

Department Poster Award
Texas A&M University
2nd Place in Department Poster Event

2024.03.01

Michael Powell Graduate Fellowship
Texas A&M University

2024.03.01

Thomas Powell 1962 Fellowship
Texas A&M University

Projects

2025.12 - 2026.02

Blackout USA
An educational tool for all backgrounds to understand the challenges of operating the electric grid.